BIOCANOPY BACKEND ARCHITECTURE & API REFERENCE

Complete Node.js, Express.js, SQLite Relational Database & External Climate API Specification

PORT: 5000 | NODE.JS ES MODULES | SQLITE DATABASE | JWT + BCRYPT | CPCB & IMD RULES

1.0 SERVER CORE & MIDDLEWARE

RUNTIME

`server/index.js`

Central application bootstrapper mounting Express router instances, CORS security, request body parsing, and server diagnostics.

- **Express HTTP Server**
  Listens on Port 5000; exposes unified REST API routes across all modules.
- **Cross-Origin Resource Sharing (CORS)**
  Strictly whitelists http://localhost:5173 with credentials: true support.
- **Body Parsing & Payload Handling**
  Built-in express.json() decodes incoming JSON telemetry and citizen submissions.
- **Database Auto-Initialization**
  Imports ./db.js on launch to verify schema tables and seed demo accounts.

GET **/api/health â€" Server Health & Diagnostics**

Returns live health status, SQLite connectivity, and active microservices (Gemini 1.5, OSRM Walking, Open-Meteo, Dispatch Queue).

2.0 RELATIONAL DATABASE (SQLITE)

PERSISTENCE

`server/db.js + database.sqlite`

Embedded ACID-compliant SQLite relational database using Node.js native DatabaseSync. Handles complete data persistence without external database servers.

TABLE: users (Citizen & Admin Accounts)
User identity, credentials, roles, and assigned authority jurisdictions.
- id (INTEGER PRIMARY KEY AUTOINCREMENT)
- name (TEXT NOT NULL) â€" Citizen full name
- email (TEXT UNIQUE NOT NULL COLLATE NOCASE) â€" Login identifier
- password (TEXT NOT NULL) â€" 10-round bcrypt hash
- role (TEXT DEFAULT 'citizen') â€" 'citizen' or 'admin'
- city (TEXT DEFAULT 'Delhi') â€" Locked assigned context (Delhi, Mumbai, Bengaluru, Ahmedabad)
- created_at (DATETIME DEFAULT CURRENT_TIMESTAMP)

TABLE: reports (Community Environmental Hazard Pins)
Geotagged crowdsourced heat and pollution reports displayed on interactive maps.
- id (INTEGER PRIMARY KEY)
- city (TEXT NOT NULL) â€" Target city name
- location_name (TEXT NOT NULL) â€" Landmark / street name
- latitude, longitude (REAL NOT NULL) â€" GPS pinpoint
- category (TEXT) â€" 'Unshaded Hotspot', 'Tree Canopy Loss', 'Dust Cloud', 'Garbage Burning'
- status (TEXT DEFAULT 'Pending Review') â€" 'Pending Review' | 'Reviewed' | 'Resolved'
- upvotes (INTEGER DEFAULT 1) â€" Community priority counter

TABLE: report_upvotes (Anti-Spam 1-Vote-Per-Citizen Registry)
Enforces atomic community voting. Prevents multiple clicks from inflating report priority.
- id (INTEGER PRIMARY KEY AUTOINCREMENT)
- report_id (INTEGER NOT NULL, FK -> reports.id)
- user_identifier (TEXT NOT NULL COLLATE NOCASE) â€" User email or session fingerprint
- created_at (DATETIME DEFAULT CURRENT_TIMESTAMP)
- CONSTRAINT: UNIQUE(report_id, user_identifier) â€" Rejects duplicate upvotes with HTTP 409

TABLE: thermal_pockets (Satellite Surface Heat Islands)
Verified high-radiation asphalt and industrial hotspots factored into pedestrian routing.
- id (INTEGER PRIMARY KEY)
- city, name (TEXT NOT NULL) â€" e.g. 'ITO Metro Gate 4 / Barakhamba Road'
- latitude, longitude (REAL NOT NULL) â€" Center coordinates
- intensity (REAL) â€" Thermal radiance score (0.0 - 1.0)
- surface_temp (REAL) â€" Peak measured pavement temperature (~48Â°C - 54Â°C)

TABLE: dispatches (Authority Squad Tasks)
Actionable authority orders for tree planting crews and mobile water misters.
- id (INTEGER PRIMARY KEY), report_id (INTEGER FK)
- task_type (TEXT) â€" 'Tree Plantation Drive' | 'Mobile Water Misting Squad'
- status (TEXT DEFAULT 'Dispatched') â€" 'Dispatched' | 'Completed'
- zone_name (TEXT), dispatched_at (DATETIME)

3.1 AUTHENTICATION & PROFILE API

/api/auth

`server/routes/auth.js`

Manages secure citizen registration, bcrypt credential encryption, JWT session tokens, and profile city-locking rules.

POST **/register**
Creates new citizen account; validates email uniqueness, hashes password (10 bcrypt rounds), and saves locked default city.

POST **/login**
Authenticates credentials via bcrypt.compare. Issues signed HMAC-SHA256 JWT bearer token (7-day validity) containing userId, role, and city.

GET **/me**
Restores active session from Authorization: Bearer <token> header. Re-verifies user existence in SQLite and returns active user state.

PUT **/profile**
Updates user profile credentials and changes assigned city in SQLite. Ensures user sees only their registered city data across the platform.

3.2 TELEMETRY & OPEN DATA API

/api/telemetry

`server/routes/telemetry.js`

Aggregates real-time weather and air pollution metrics across Indian metros, calculating official CPCB NAQI and asphalt thermal radiance.

GET **/live?city=:name**
Queries Open-Meteo & aqi.in CAAQMS ground stations. Returns air temp, humidity, wind, PM2.5, PM10, Heat Index, and Asphalt Radiant Temp (3-min cache).

- **CPCB NAQI Sub-Index Calculator**
  Custom mathematical engine converting raw PM2.5/PM10 concentrations into India's official National Air Quality Index (0-500 scale) using legal breakpoints.
- **Asphalt Radiant Heat Gain Model**
  Models unshaded asphalt thermal absorption: ground surface radiates 7Â°C - 10Â°C hotter than ambient air, driving urban heat island calculations.

GET **/historical**
Supplies 24-hour and 7-day diurnal curves for temperature, humidity, and AQI fluctuations, powering interactive Recharts area graphs.

GET **/export?format=csv|json**
Data streaming pipeline providing instant downloads of verified environmental telemetry for university students and climate analysts.

3.3 COOL-ROUTE NAVIGATION ENGINE

/api/routes

`server/routes/routes.js`

Pedestrian heat-resilience routing engine combining OpenStreetMap road geometry with satellite heat islands and tree canopy coverage.

POST **/calculate**
Receives [origin, destination] coordinates. Queries OSRM Walking API for real street geometry and pedestrian maneuver instructions.

- **Thermal Penalty & Canopy Shade Bonus**
  Dual-route comparison: standard path reflects bare asphalt, while Cool-Route favors green-canopied corridors with up to -34% radiant heat reduction.
- **Hazard Pin Collision Avoidance**
  Dynamically penalizes pedestrian paths passing within 150m of active citizen hazard pins (e.g. unshaded asphalt or active dust clouds).

GET **/thermal-pockets?city=:name**
Fetches urban heat island coordinates [lat, lng, intensity] rendered on Leaflet thermal heatmaps.

3.4 CITIZEN HAZARD REPORTING API

/api/reports

`server/routes/reports.js`

Crowdsourced environmental reporting platform enabling citizens to flag heat emergencies, unshaded stretches, and dust clouds.

GET **/?city=:name&status=:status**
Returns list of citizen reports with GPS coordinates, upvotes, photo URLs, and review statuses for community display.

POST **/**
Submits new geotagged hazard report with category, location name, description, photo link, and defaults to 'Pending Review'.

POST **/:id/upvote**
Enforces strict 1-vote-per-user system using report_upvotes. Rejects repeated clicks with HTTP 409 Conflict, completely eliminating vote spam.

3.5 AI CLIMATE & HEALTH ADVISOR

/api/advisor

`server/routes/advisor.js`

Hybrid medical intelligence system combining CPCB/IMD validated rules with Google Gemini 1.5 Flash for trilingual health directives.

POST **/generate**
Generates clinical heatwave alerts, hydration targets, and safe transit windows in English, Hindi, or Marathi based on live ambient telemetry.

- **CPCB & IMD Validated Rules Engine v2.4**
  Clinical matrix categorizing physiological heat stress (Extreme Heatwave >=42Â°C, Peak Solar Heat 37-41Â°C, Moderate Warm 32-36Â°C, Comfortable <32Â°C).

POST **/chat**
Powers the interactive floating AI Climate Copilot. Evaluates user questions against live temperature data with strict medical heat-safety guardrails.

3.6 AUTHORITY COMMAND & DISPATCHES

/api/admin

`server/routes/admin.js + dispatches.js`

Authority management console enabling field officers to triage citizen hazard flags, manage users, and dispatch tree planting crews.

GET **/stats**
Aggregates city-wide KPIs: total reports, pending triage, resolved hotspots, active dispatches, and registered citizens.

PATCH **/reports/:id/status**
Updates report review lifecycle: 'Pending Review' -> 'Reviewed' -> 'Dispatched' -> 'Resolved'.

POST **/dispatches**
Creates authority work orders (Tree Plantation Squad, Water Misting Unit); notifies field teams and residents.

4.0 CONNECTED ONLINE PUBLIC & CLOUD APIS

EXTERNAL SERVICES

Global Cloud Integrations

- **Open-Meteo Weather API**
  `GET https://api.open-meteo.com/v1/forecast`
  Provides ambient air temperature, relative humidity, apparent feels-like temp, and wind velocity.
- **Open-Meteo Air Quality API**
  `GET https://air-quality-api.open-meteo.com/v1/air-quality`
  Provides satellite atmospheric chemical telemetry for real-time PM2.5 and PM10 concentrations.
- **AQI.in Ground Station Scraper**
  `GET https://www.aqi.in/in/dashboard/india/...`
  Cross-references CAAQMS physical ground stations (CPCB & MPCB) for ground truth validation.
- **Project OSRM Walking API**
  `GET https://router.project-osrm.org/route/v1/walking/...`
  Computes true pedestrian street routing, road network distances, and step-by-step turn maneuvers.
- **Google Gemini 1.5 Flash API**
  `POST https://generativelanguage.googleapis.com/...`
  Cloud Generative AI providing contextual climate copilot responses with medical guardrails.